Intermediate Elective

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Set Up

Packages

```
# general use
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```
library(snakecase)
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
# file organization
library(here)
```

```
here() starts at /Users/zeldareif/Desktop/github/intermediate-elective
```

```r
# reading .xlsx files
library(readxl)

# data visualization
library(naniar)
library(patchwork)
library(NatParksPalettes)
library(gghighlight)
library(paletteer)
library(ggplot2)
```

Data

```r
# vegetation
veg <- read_csv(here("data", "veg copy.csv"))
```

```
Rows: 27036 Columns: 17
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (10): monitors, transect_name, vp_axis, site, transect_side, vp_zone, q...
dbl   (6): x1, transect_length, num_quad, transect_distance, percent_cover, ...
date  (1): date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# veg metadata
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

```
Rows: 257 Columns: 4
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): year_pool, transect_name
```

```
dbl (2): year, num_quad

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# taxonomic info
taxon_list <- read_csv(here("data","taxon_list.csv"))
```

```
Rows: 50 Columns: 15
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (15): verbatim_name, taxonRank, scientificName, kingdom, Phylum, Class, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site info
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")
```

Problems

1. Explain your inspiration

I was inspired by Steven Ponce's "Evolution of MTA Art Materials (1980-2020)" because the figure's characteristics would allow me to show total percent cover of multiple vernal pool vegetation species through time, while highlighting one species per faceted plot.

This figure makes sense for the vernal pool vegetation data set because it allows for a clear comparison of total percent cover between different species with distinct lines.

The plot will be faceted by vegetation species, the x axis will be time by year and the y axis will be total percent cover. Each plotted line will represent a different vernal pool vegetation species, and the species of interest's line will be highlighted with a different color to stand out.

2.

Sketch what your figure would look like following your inspiration.

Insert that sketch (as a photo, screenshot, or otherwise) into your document.

3. Code your figure

```
taxon_list_clean <- taxon_list |>
  # convert taxon names to snakecase
  mutate(verbatim_name = to_any_case(verbatim_name,
  case = "snake"))

# creating new object
ncos_sites <- sites |>
# filtering to only include ncos quarterly sampling sites
filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

water_quality_clean <- water_quality |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  unite("date_site", date, site, remove = FALSE) |>
  group_by(date, site, date_site) |>
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  ungroup() |>
  filter(date_site != "2022-08-12_NVBR")
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by date, site, and date_site.
i Output is grouped by date and site.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(date, site, date_site))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```r
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5
  # liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(
  select(water_quality_clean,
         date_site,
         med_ph,
         med_sal,
         med_do),
         by = "date_site") |>
```

```
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
        site == "NVBR" ~ "North of Venoco Bridge",
        site == "NEC" ~ "South of Venoco Bridge",
        site == "NMC" ~ "Main Channel",
        site == "NPB" ~ "South of Phelps Bridge",
        site == "NPB1" ~ "North of Phelps Bridge",
        site == "NPB2" ~ "Phelps Road",
        site == "NWP" ~ "West Pond",
        site == "NDC" ~ "Devereux Creek")) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
= TRUE),
  site_full = fct_rev(site_full))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by date, site, and taxon.
i Output is grouped by date and site.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(date, site, taxon))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

```
# create new object from aq_ins_clean
ost_abundance <- aq_ins_clean |>
  # filter to only ostracods
  filter(taxon == "ostracod") |>
  # remove missing abundance/L values
  filter(!is.na(ave_lit)) |>
  # group by site
  group_by(site_full) |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
```

```
  # ungroup data frame
  ungroup()
```

```
y_max <- max(ost_abundance$log_ave_lit, na.rm = TRUE)
ring_log <- pretty(c(0, y_max), n = 4)

ostracods_plot <- ggplot(
  data = ost_abundance,
  mapping = aes(
    x = site_full,
    y = log_ave_lit
  )
) +

  geom_hline(
    yintercept = ring_log,
    color = "lightgray"
  ) +

  geom_vline(
    xintercept = 1:length(unique(ost_abundance$site_full)),
    linetype = "dashed",
    color = "gray30"
  ) +

  geom_jitter(
    height = 0,
    width = 0.1,
    shape = 21
  ) +

  stat_summary(
    geom = "point",
    fun = "median",
    size = 2,
    color = "lightblue"
  ) +

  scale_x_discrete(labels = \(x) str_wrap(x, 15)) +

  scale_y_continuous(
    breaks = ring_log,
```

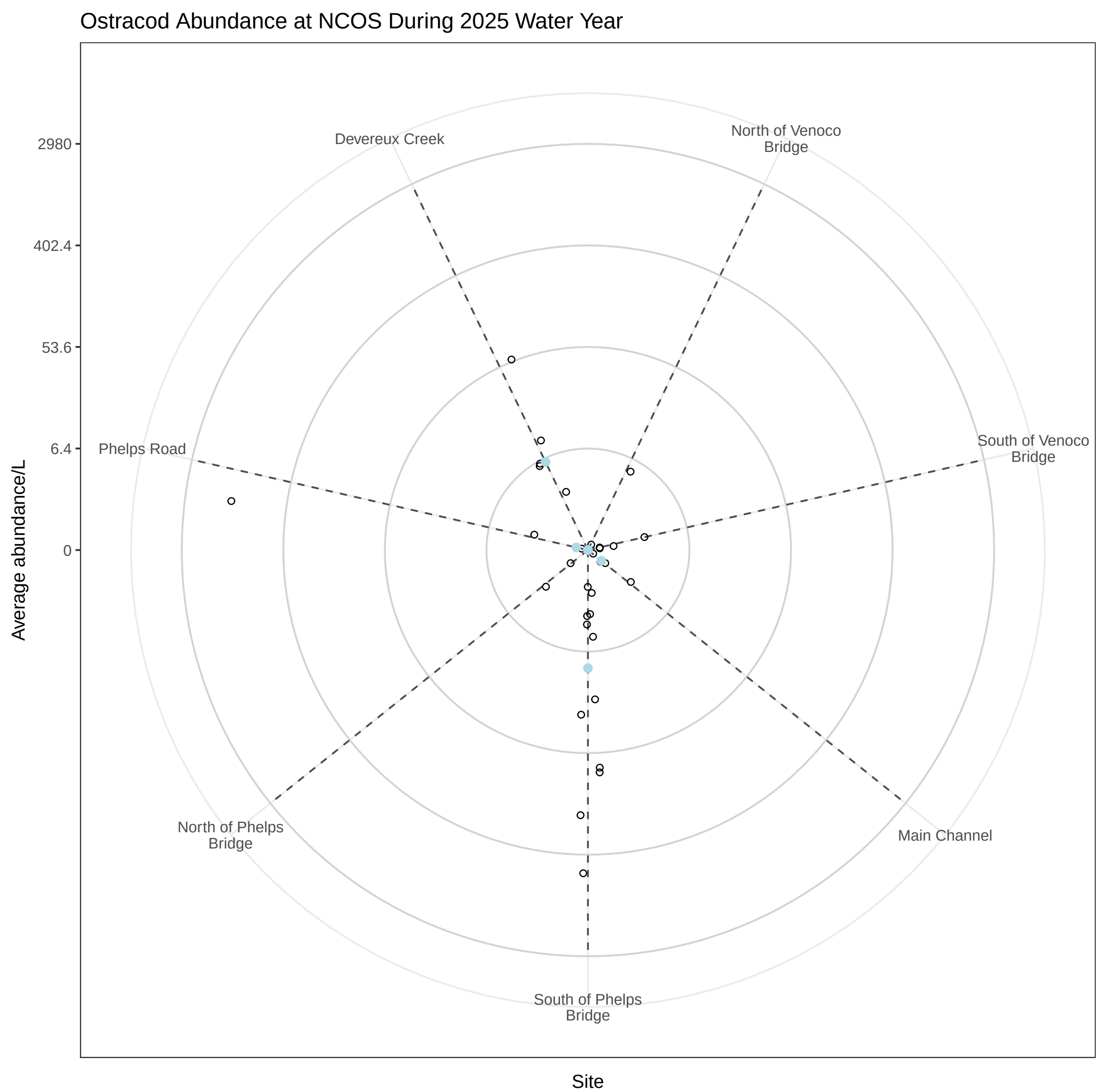
```
    labels = round(exp(ring_log) - 1, 1)
  ) +

  labs(
    x = "Site",
    y = "Average abundance/L",
    title = "Ostracod Abundance at NCOS During 2025 Water Year"
  ) +

  coord_polar() +

  theme_bw()

ostracods_plot
```




```
ostracods_plot <- ggplot(
  data = ost_abundance,
  mapping = aes(
    x = site_full,
    y = log_ave_lit
  )
) +

  geom_hline(
```

```
    yintercept = ring_log,
    color = "lightgray"
  ) +

  geom_vline(
    xintercept = 1:length(unique(ost_abundance$site_full)),
    linetype = "dashed",
    color = "gray30"
  ) +

  geom_jitter(
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    geom = "point",
    fun = "median",
    size = 2,
    color = "lightblue"
  ) +

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    x = "Site",
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  ) +

  coord_polar() +

  theme_bw()

ostracods_plot
```

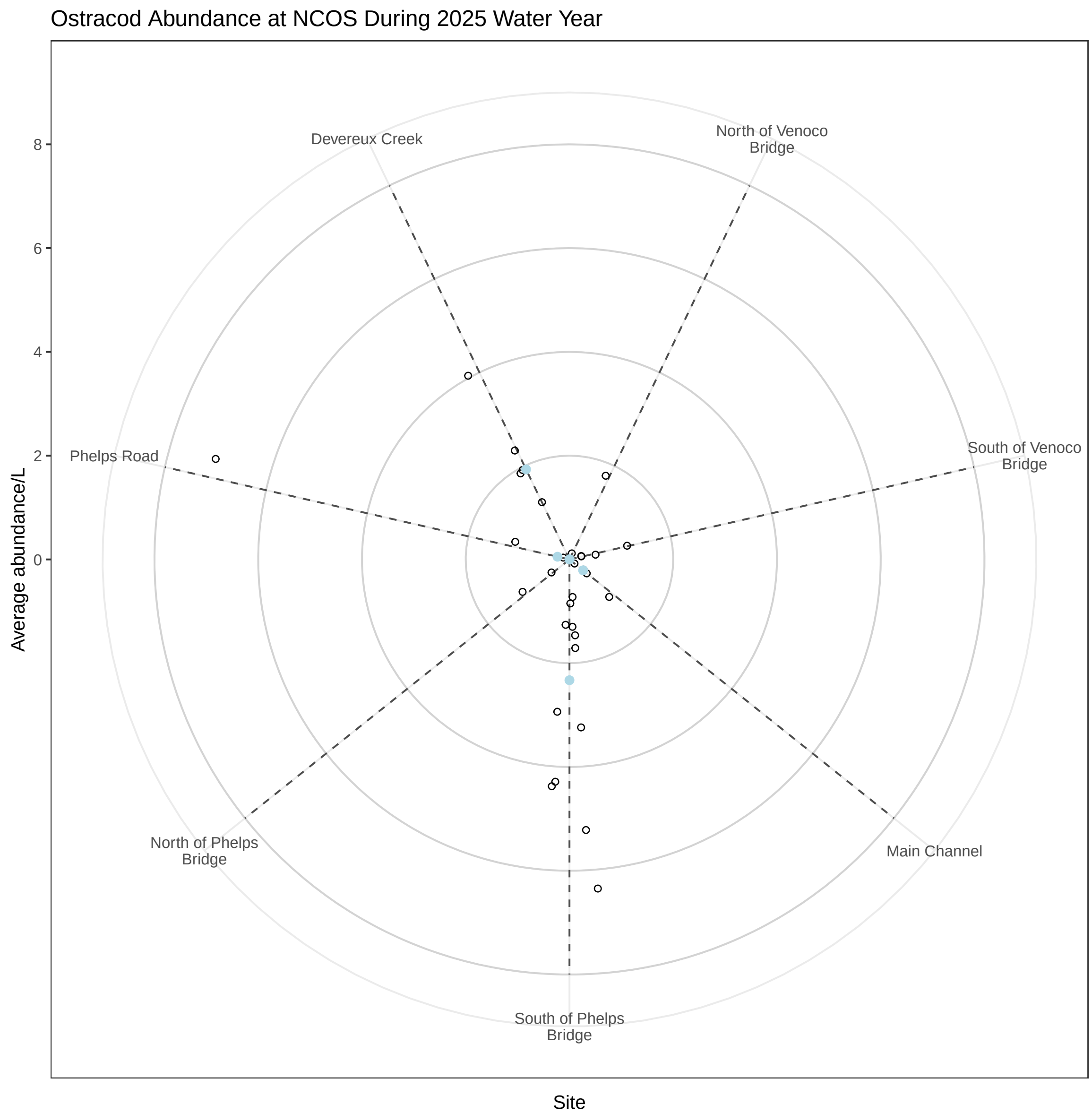

4. Describe your process

In 1-3 sentences each, describe:

- your process of using someone else's code to create a visualization
- challenges you encountered as you made your visualization
- how your visualization differs from visualizations you made in class, for assignments, or for a previous elective